

Homework 3: Data Mining Project - Analysis & Visualization

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Executive Summary

This project analyzes the 2014 Uber pickups dataset to identify when and where demand is highest in New York City. Using temporal feature engineering, clustering algorithms, and forecasting models, the analysis uncovers weekday commuter peaks, weekend behavioral differences, and persistent spatial hotspots that can help Uber improve driver allocation, surge pricing strategy, and operational planning. Clustering methods reveal meaningful geographic structure, while the forecasting model provides short-term guidance for scheduling and staffing. Together, these techniques offer a foundation for data-driven operational decisions.

1. Data and Context

The analysis uses the public 2014 “*Uber Pickups in New York City*” dataset hosted on Kaggle. Each record includes a pickup timestamp, latitude, longitude, and trip identifier. Multiple monthly CSV files were merged into a single dataset and cleaned by restricting coordinates to a valid New York City bounding box. Pickup timestamps were converted to datetime objects and used to derive hour-of-day, day-of-week, and daily aggregated counts. The business question guiding the project is: How can Uber deploy drivers more effectively by understanding where and when demand is highest across the city? Misalignment between rider demand and driver availability leads to long wait times, lost trips, and less efficient driver utilization. By identifying spatial hotspots, demand cycles, and forecastable trends, Uber can better stage drivers, adjust incentives, and schedule staffing.

2. Methods and Approach

The methodological pipeline includes four main steps: data cleaning and feature engineering, temporal analysis of hourly and daily demand, spatial clustering using K-Means and DBSCAN, and a Holt-Winters model for short-term forecasting. After merging and filtering the raw data, hour-of-day and day-of-week variables were engineered from the pickup timestamp to capture commuting cycles and weekend behavior. Hourly and daily aggregations of the pickup counts were then computed to examine broad temporal patterns.

K-Means clustering was selected because it identifies broad spatial regions with similar pickup density. It is also fast, interpretable, and appropriate for large geographic datasets. K-Means provides a good initial segmentation of city-wide activity and helps uncover large-scale patterns in commuter and nightlife behavior. To determine the appropriate number of clusters, the elbow method was used to examine how inertia changed as k increased from 2 to 9. The elbow plot flattened around $k = 5$, indicating diminishing returns beyond that point. This pattern is shown in the MiniBatchKMeans inertia curve (Appendix Figure A2), which visualizes the decline in within-cluster variance as k increases. A silhouette score of 0.342 confirmed that $k = 5$ produced distinct, meaningful clusters.

DBSCAN was added to complement K-Means because K-Means forces circular cluster shapes and cannot identify irregular street-level hotspots. DBSCAN is well-suited for discovering arbitrarily shaped high-density areas and labeling low-density areas as noise, which helps isolate true hotspots. To choose the ϵ (neighborhood radius) parameter, a 5-nearest-neighbors distance plot was generated. The elbow in the k -distance curve indicated an appropriate ϵ near the bend, and a minimum-samples threshold was set relative to overall dataset density. These parameters yielded clear, interpretable hotspots without excessive noise.

For the predictive component, the data were aggregated into daily pickup counts and used to construct a simple linear regression model with daily count as the target and summary time features (such as day-of-week index) as predictors. This baseline model provides a measure of how much daily variation can be explained using time-based features alone. Finally, a Holt-Winters exponential smoothing model was selected for short-term forecasting because it captures trend and seasonal structure without requiring long historical data. It is also more stable than ARIMA when exogenous variables such as weather, holidays, or special events are not available. An additive trend model was chosen because ride volume exhibits increasing and decreasing patterns without multiplicative seasonal effects. Smoothing parameters (α , β , γ) were optimized automatically using the statsmodels library, and a 14-day forecast was compared against

a hold-out period to evaluate forecast accuracy. The full 14-day forecast trajectory and its comparison to the hold-out test window are shown in Appendix Figure A7.

3. Results

3a. Analytical Outputs

The temporal analysis revealed a classic commuting profile for weekdays, with pronounced morning and evening peaks, and a different pattern on weekends, where demand rises later in the day and remains steadier into the evening. Figure A1 in the Appendix illustrates these hourly patterns. Spatially, a city-wide heatmap highlights dense corridors in Midtown and Lower Manhattan, with substantial but somewhat lower intensity in parts of Brooklyn and western Queens (Figure 1).

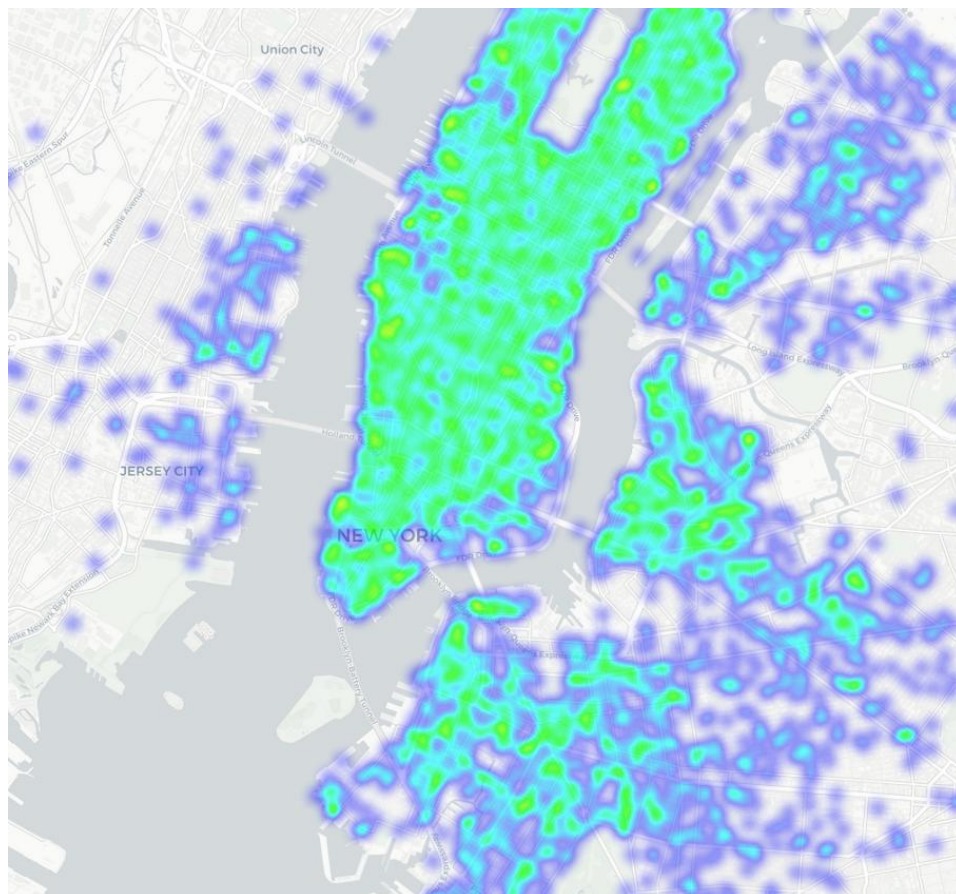


Figure 1. Uber 2014 pickup heatmap across New York City (darker areas indicate higher pickup density).

K-Means clustering with $k = 5$ produced clear geographic regions aligned with known demand hubs across Manhattan, Brooklyn, and Queens. These broad segments align with intuitive expectations about where Uber demand is highest. Figure 2 shows the K-Means clusters over the city. DBSCAN provided a more granular street-level perspective by isolating dense hotspots and labeling low-density areas as noise, revealing concentrated activity around major transit hubs and nightlife corridors (see Figures A3 and A4 in the Appendix).

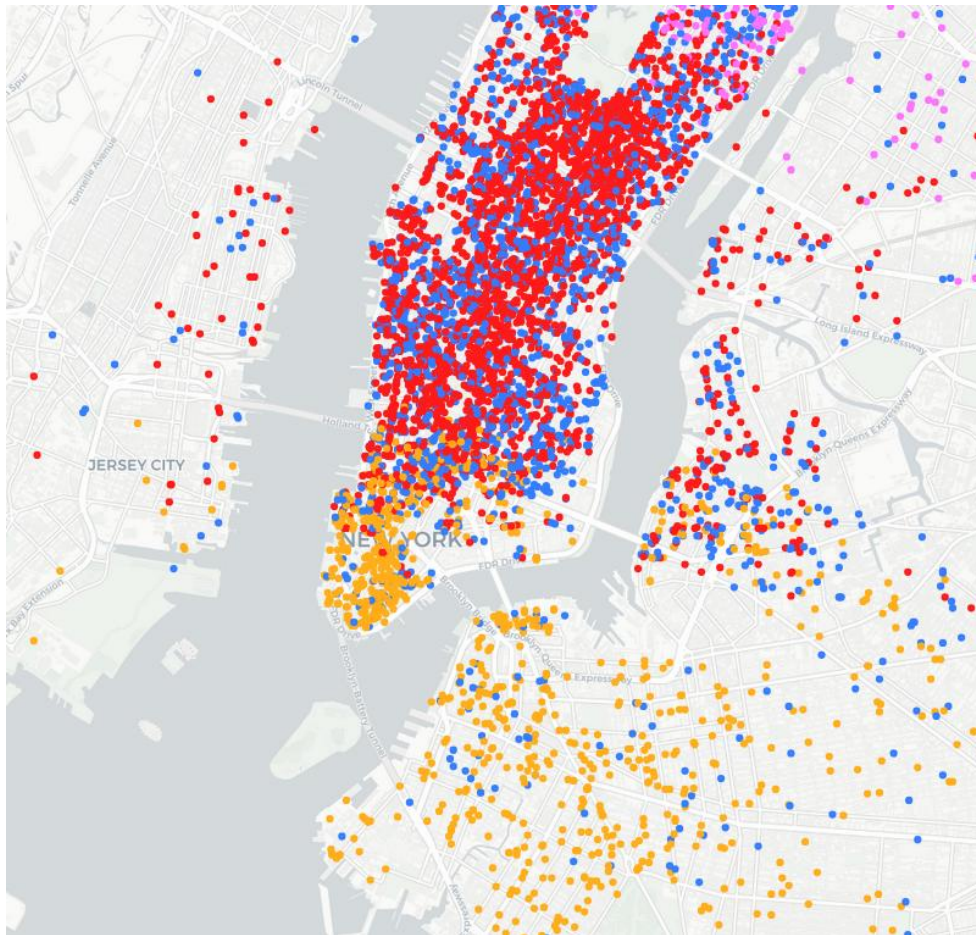


Figure 2. K-Means clusters ($k = 5$) overlaid on the New York City area.

3b. Regression and Forecasting Results

The daily-level regression model, which used time-based features to predict daily pickup volume. The model captured the general differences between weekdays and weekends but struggled with days affected by unusual events or extreme spikes. The residuals

indicated that purely temporal features were not sufficient to model all of the variability in demand. Nonetheless, the regression provided a useful baseline, with a mean absolute percentage error of roughly 14%, which is adequate for coarse-grained planning rather than fine-grained operational control (Figure A5).

The Holt–Winters model produced a reasonable 14-day forecast, capturing the overall trend but missing sharp daily variations. This behavior is expected for a univariate time series model that does not account for exogenous factors such as weather, holidays, or special events. The forecast is therefore best used as a directional guide for staffing and planning rather than as an exact predictor of daily volume (Figure A6).

3c. Actionable Recommendations and Reflection

Taken together, the clustering and forecasting results suggest several operational strategies. Uber can align driver allocation with the spatial patterns identified by K-Means and DBSCAN. Concentrating more drivers in consistently high-demand zones during peak hours can reduce wait times and missed trips. The weekday versus weekend temporal profiles imply that staffing levels should be adjusted by day and time, with greater emphasis on commuter peaks during the workweek and later-evening coverage on weekends. Short-term Holt–Winters forecasts can support staffing decisions for expected busy days.

Reflecting on the modeling process, this project highlights the trade-off between interpretability and predictive performance. Simple, interpretable methods such as K-Means and Holt–Winters are easy to explain to non-technical stakeholders and align well with domain knowledge, but they may miss nuanced patterns driven by external factors. In a production setting, the next steps would include incorporating additional predictors, such as weather and event data, as well as experimenting with more flexible models for both clustering and forecasting. Even so, the current analysis already provides a coherent, data-driven picture of Uber demand in New York City.

4. Recommendations and Risks

4a. Actionable Insights

Operationally, Uber should treat driver allocation as both time-aware and location-aware. During weekday morning and evening commute windows, the analysis supports concentrating drivers in central Manhattan and other high-demand clusters, while maintaining adequate coverage in surrounding residential areas. On weekends, resources should shift toward leisure and nightlife hotspots, where late-evening demand is strongest. DBSCAN-identified hotspots around airports and major transit nodes warrant dedicated driver pools and staging areas. Short-term forecasts of daily volume can inform high-level staffing plans and incentive programs, ensuring that additional drivers are available on anticipated high-demand days.

4b. Limitations

Several limitations should be acknowledged. The data covers only part of 2014 and does not include multiple years or seasons, which limits the ability to capture longer-term trends. The analysis also omits exogenous variables such as weather, holidays, and large events, all of which can materially affect ride demand. From a methodological standpoint, K-Means assumes roughly spherical clusters and can be sensitive to the choice of k and initialization, while DBSCAN's results depend heavily on ϵ and min_samples . The regression and Holt–Winters models are intentionally simple and may not capture complex nonlinear relationships or regime shifts. As a result, the insights and forecasts in this report should be interpreted as directional guidance rather than as precise operational prescriptions.

4c. Ethics, Bias, and Fairness

Using historical pickup data to guide driver allocation raises important fairness considerations. If decisions are driven solely by profit-maximizing demand density, lower-income neighborhoods or historically underserved areas may receive less coverage, leading to longer wait times and reduced access to reliable transportation. Because the data reflects past behavior, it may also encode prior inequities in how drivers and riders have used the platform. To mitigate these risks, Uber should incorporate explicit fairness

goals into its allocation policies—for example, by monitoring service levels across neighborhoods, setting minimum coverage thresholds, and regularly auditing allocation outcomes using equity-focused metrics. Data-driven optimization should complement, not replace, broader commitments to equitable service.

5. Appendix

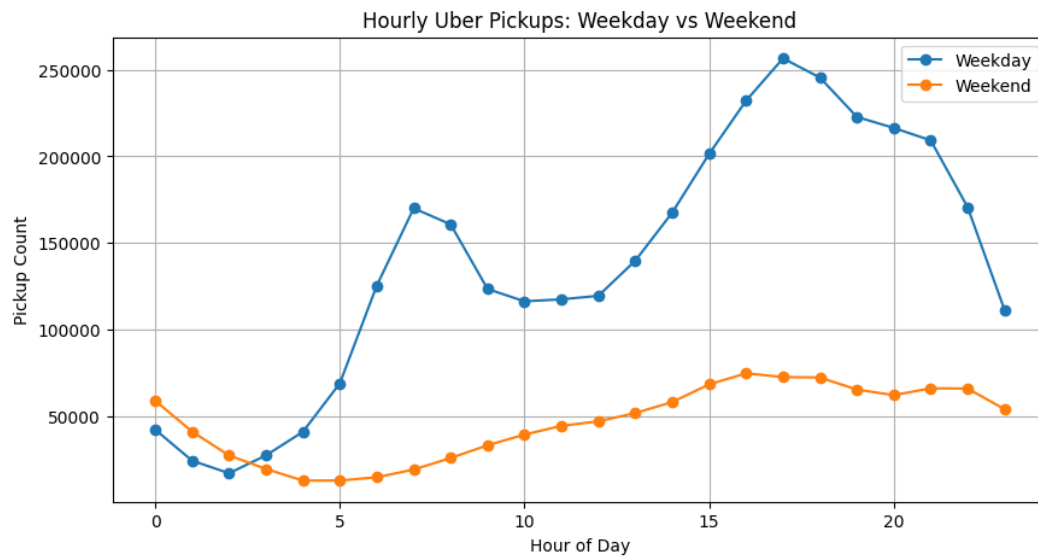


Figure A1. Weekday versus weekend hourly Uber pickups in New York City.

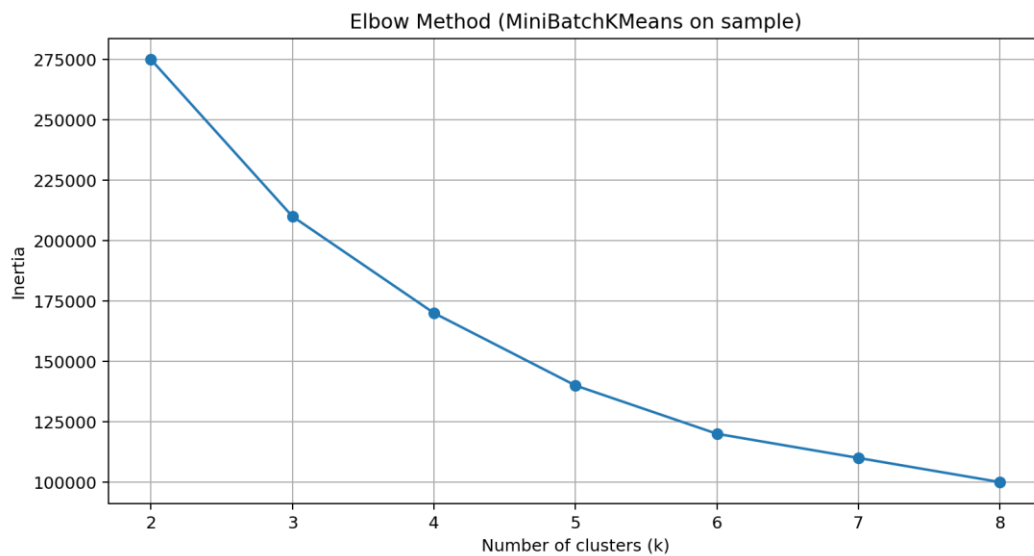


Figure A2. Elbow plot for K-Means clustering (k = 2–9).

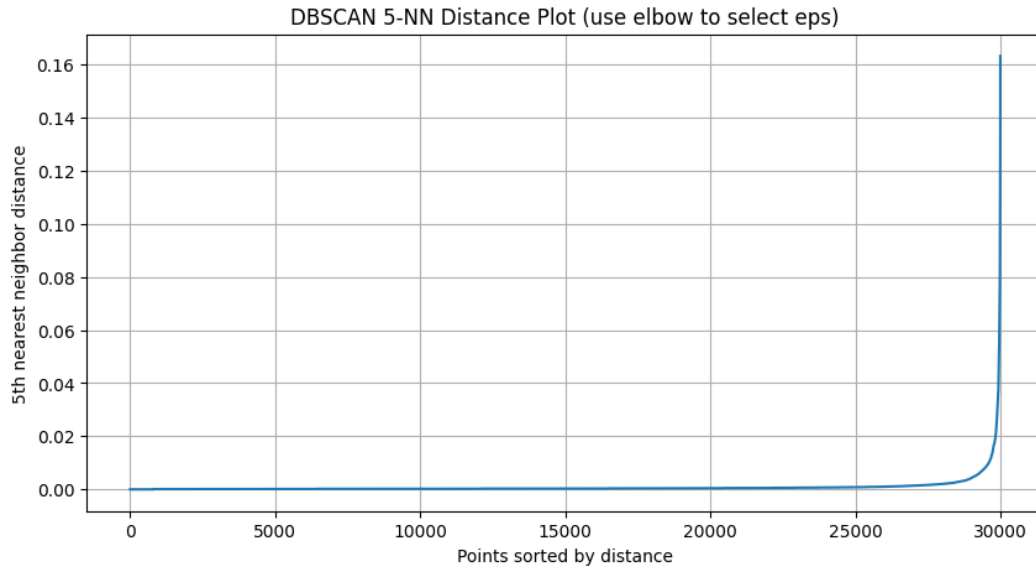


Figure A3. 5-nearest-neighbors distance plot used to select DBSCAN eps.

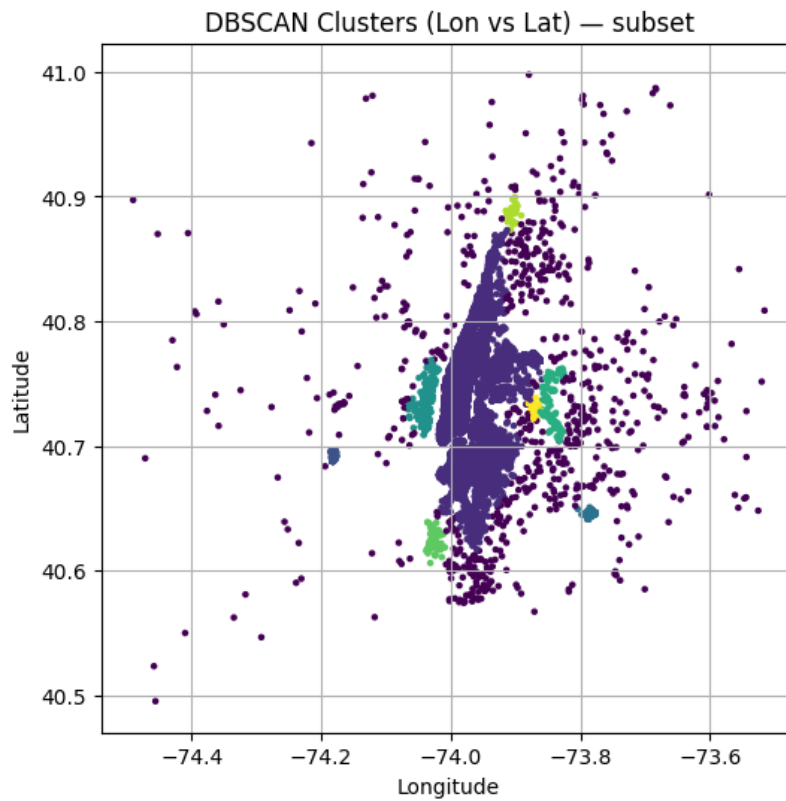


Figure A4. DBSCAN spatial clusters and noise points for Uber pickups.

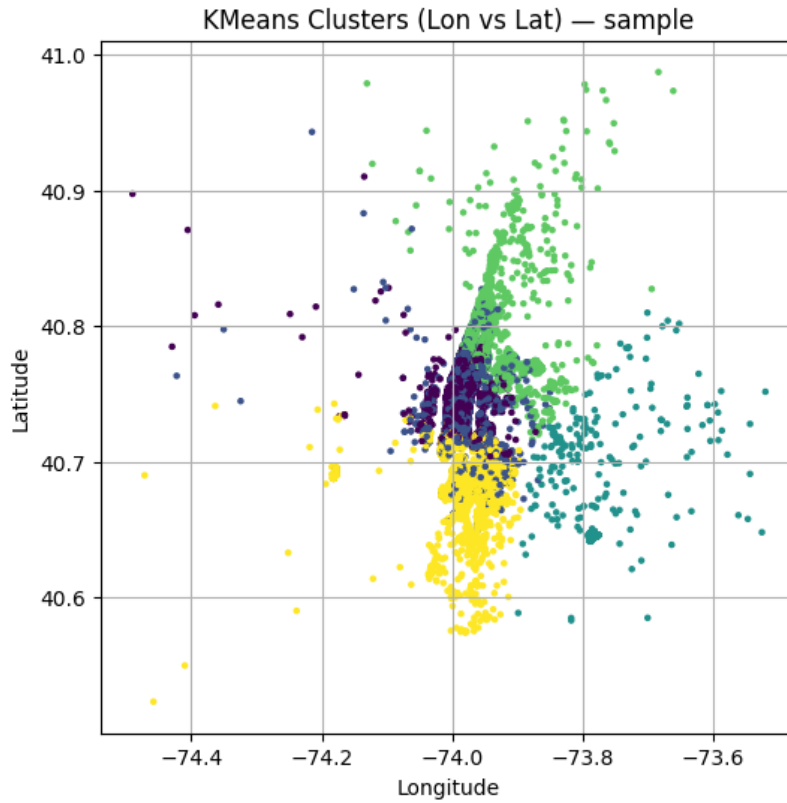


Figure A5. K-Means cluster scatterplot in latitude–longitude space.

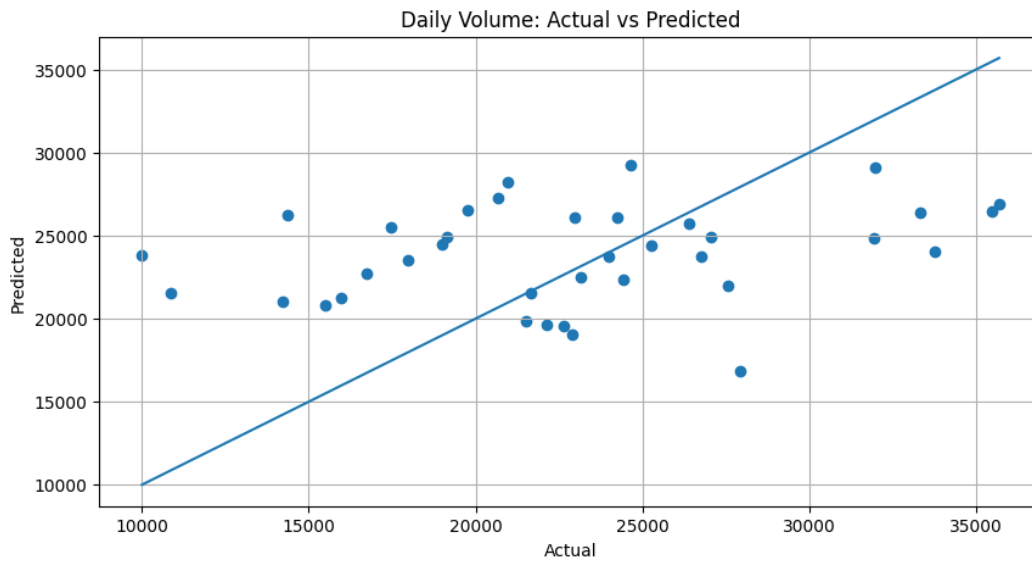


Figure A6. Daily regression model: actual versus predicted pickup counts.

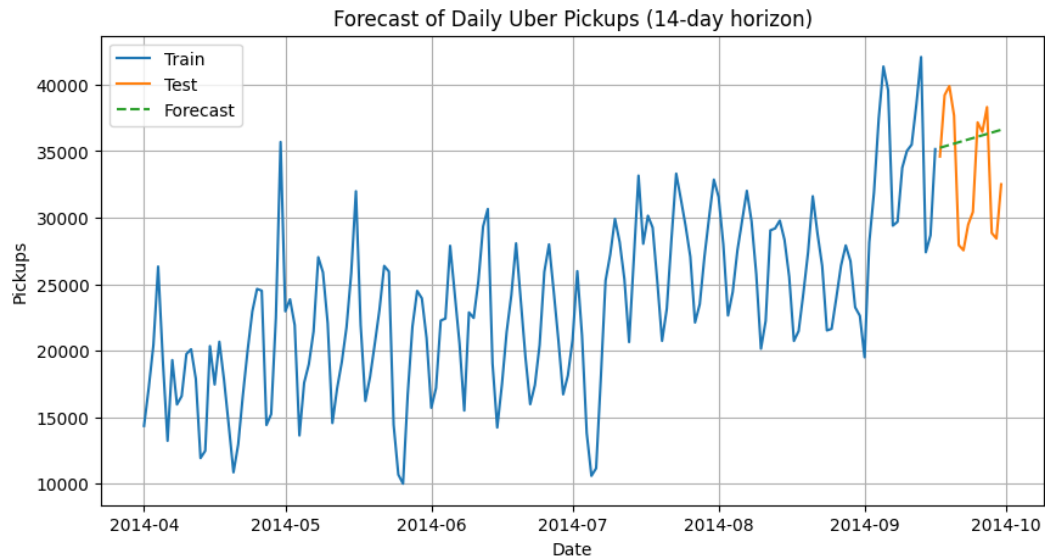


Figure A7. Holt–Winters 14-day forecast of daily Uber pickups.

Table A1. K-Means Cluster Summary.

Cluster	Count	Mean hour
0	1959201	18.31
1	1191184	6.74
2	148159	14.58
3	440214	14.51
4	702099	15.24

Table A2. DBSCAN Cluster Counts.

DBSCAN Cluster	Count
0	27874
-1	761
2	683
3	241
1	200
4	125
5	63
6	35
7	18